

實作的三個 function（包含註解）：

1. get_balance

```
def get_balance(self, account):
    balance = 0
    # -----TODO-----
    ## Traverse the blockchain to get the balance
    for block in self.chain:
        # Check miner reward
        miner = False

        if block.miner == account: # 如果區塊的礦工是指定的帳戶，則標記為礦工。
            miner = True
            balance += block.miner_rewards
        for transaction in block.transactions:
            if miner: # 如果是礦工，增加交易手續費到餘額。
                balance += transaction.fee

            if transaction.sender == account: # 如果交易的發送者是指定的帳戶，從餘額中減去轉出金額和交易手續費。
                balance -= transaction.amounts
                balance -= transaction.fee

            elif transaction.receiver == account: # 如果交易的接收者是指定的帳戶，增加接收金額到餘額。
                balance += transaction.amounts

    # Add the test token that the customer had already applied
    for i in self.address_table:
        if account == i[0]:
            balance += i[1]
            break

    return balance
```

2. initialize_transaction(in miner.py)

```
def initialize_transaction(self, sender, receiver, amount, fee, message):
    # -----TODO-----
    # 使用get_balance確認交易起方有足夠的帳戶餘完成這筆交易!
    ## 若帳戶餘額不足請回傳 "Balance not enough!"
    if self.get_balance(sender) < amount + fee:
        print("Balance not enough!")
        return False
    ## 若帳戶餘額足夠則回傳一個Transaction物件
    new_transaction = Transaction(sender, receiver, amount, fee, message)
    return new_transaction
```

3. initialize_transaction(in customer.py)

```
def initialize_transaction(sender, receiver, amount, fee, message):
    # -----ToDo-----
    ## Same to the work in miner.py , but no need to check balance !!!
    ## 請回傳一個Transaction物件
    new_transaction = Transaction(sender, receiver, amount, fee, message)
    return new_transaction
```

Snapshots:

1. Start mining

```
$ python miner.py
Address: MEgCQQC0ipKBqL9K2330Enb4XCs1SCdPfRV062CRztKDCiWZ3EBf0ZvpG2IJ8nc7HuQ7PiD
ovz7NuWqQAbxvq+wkQ771AgMBAAE=
Create genesis block...
In interrupt_control function : input value 0 to interrupt current work on mining
new block. Hash found: 000004aeda0e8da814993a1f1cac42bb851afbdf @ difficulty 5,
time cost: 2.85074s
balance amount: 10
Hash found: 00000fabb1882399c028021ded4d03a8bc44c210 @ difficulty 5, time cost:
0.18364s
balance amount: 20
Hash found: 00000a2f69f969dbcbcb8561c24e65797f80f0d5c @ difficulty 5, time cost:
3.93013s
balance amount: 30
Hash found: 00000e186ae0b10eda5f5d317c595ae2275950 @ difficulty 5, time cost:
0.70739s
balance amount: 40
Connected by: ('127.0.0.1', 52713)
Hash found: 000005c6815d20da10daafdb7bea3998d00169cf @ difficulty 5, time cost:
2.80454s
balance amount: 50
Hash found: 0000053ba3e18b6382d01aec271571ae4122d12f @ difficulty 5, time cost:
2.4853s
```

2. 用 command 1 生成 address

```
Address: MEgCQQCW4GVfHAZTUJjAw7LRAQXDL1BGF9Exzk08vtJ5YvUSrvZ+JqsMbQtWTTjpXbNEQ1G
d6GR0Go+acBTBInIC7HTTAgMBAAE=
Private key: MIIBPAIBAAJBABJbgZV8cBlNQmMDDuVEBBcMvUEYX0TH0Q7y+0nli9RKu9n4mqwxtC1Z
N00lds0RDUZ3oZE4aj5pwFMEicgLsdNMCAwEAAQJAHsekpqSW/cRBhYHoeogU3t72J3aQbPLBiu37Dly
Lpqneoh0ie6/odGH5sb2JFa2suVyHonVsVdKqtxPUL5BLkQIjA0vS0fVKu3SfTqxdm705+5kc+WaZxy8
j/wkifcAndjbbn18CHWcJyQFCfqi9mYnnD+BTmejuoNENLoJjI7QTACygww0CInSi2tF1Wu29dU3vJnl
n/Tani/rhCBEavqMn+gY1/LP3oAUCHkjcq+3L8P9f5aRgQLiQWLqH2lhV33WBNaPaSQA/pQIjAL//5SE
zsVkayaK/VwShRj7nKldm//txQz+Qp++J8kVJlWU=
Successfully send the address to miner.
Update local address table.
Node 1 : MEgCQQC01hJ790jKPATtZgpEK6JpMsw6/3s22AW30qI3bNTOHAcMEjjXzBkM/Lc0dE504Y
6g2EKQnVbee2vf+6Q8oizfAgMBAAE=
```

3. 用 command 2 取得 balance

```
Command: 2
Address: MEgCQQCW4GVfHAZTUJjAw7LRAQXDL1BGF9Exzk08vtJ5YvUSrvZ+JqsMbQtWTTjpXbNEQ1G
d6GR0Go+acBTBInIC7HTTAgMBAAE=
Balance = 0
```

4. 用 command 5 取得一些 token (第一張圖 sender、第二張圖 receiver)

```
Command: 5
Address: MEgCQQCW4GVfHAZTUJjAw7LRAQXDL1BGF9Exzk08vtJ5YvUSrvZ+JqsMbQtWTTjpXbNEQ1G
d6GR0Go+acBTBInIC7HTTAgMBAAE=
How many tokens you are going to apply : 20
```

```
Command: 5
Address: MEgCQQCUvg1LRdo7d4Lc3vZmJpjLSnB7MwotakJGakDaGSF2Cm24o5QkgNUeMB45sENhBB6
pk6CLVS3PLAhetDnR+q+LAgMBAAE=
How many tokens you are going to apply : 15
```

5. 用 command 2 檢查 sender 的 balance 是否有變動

```
Command: 2
Address: MEgCQQCW4GVfHAZTUJjAw7lRAQXDL1BGF9Exzk08vtJ5YvUSrvZ+JqsMbQtWTTjpXbNEQ1G
d6GR0Go+acBTBInIC7HTTAgMBAAE=
Balance = 20
```

6. 用 command 2 檢查 receiver 的 balance 是否有變動

```
Command: 2
Address: MEgCQQCUvg1LRdo7d4Lc3vZmJpjLSnB7MwotakJGakDaGSF2Cm24o5QkgNUeMB45sENhBB6
pk6CLVS3PLAhetDnR+q+LAgMBAAE=
Balance = 15
```

7. 用 command 3 讓兩個 customer 做交易

```
Command: 3
Address: MEgCQQCW4GVfHAZTUJjAw7lRAQXDL1BGF9Exzk08vtJ5YvUSrvZ+JqsMbQtWTTjpXbNEQ1G
d6GR0Go+acBTBInIC7HTTAgMBAAE=
Private_key: MIIBPAIABAJBAJbgZV8cBlNqMDDuVEBBcMvUEYX0THOQ7y+0nli9RKu9n4mqwxtC1Z
N00lds0RDUZ3oZE4aj5pwFMEicgLsdNMCAwEAAQJAHsekpqSW/cRBhYHoeogU3t72J3aQbPLBiu37Dly
Lpqneoh0ie6/odGH5sb2JFa2suVyHonVsVdKqtXPUL5BLkQIjA0vS0fVKu3SfTqxdm705+5kc+WaZxy8
j/wkifcAndjbbn18CHwCjyQFCfqi9mYnnD+BTmejuoNENLoJjI7QTACyggw0CInSi2tF1Wu29dU3vJn1
n/Tani/rhCBEavqMn+gY1/lP3oAUCHkjcq+3L8P9f5aRgQLiQWLqH2lhV33WBNaPaSQA/pQIjAL//5SE
zsVkayaK/VwShRj7nKldm//txQz+Qp++J8kVJlWU=
Receiver: MEgCQQCUvg1LRdo7d4Lc3vZmJpjLSnB7MwotakJGakDaGSF2Cm24o5QkgNUeMB45sENhBB
6pk6CLVS3PLAhetDnR+q+LAgMBAAE=
Amount: 5
Fee: 1
Comment: 123!
True
```

8. 交易完用 command 2 檢查 sender 的 balance 是否有變動

```
Command: 2
Address: MEgCQQCW4GVfHAZTUJjAw7lRAQXDL1BGF9Exzk08vtJ5YvUSrvZ+JqsMbQtWTTjpXbNEQ1G
d6GR0Go+acBTBInIC7HTTAgMBAAE=
Balance = 14
```

9. 交易完用 command 2 檢查 sender 的 balance 是否有變動

```
Command: 2
Address: MEgCQQCUvg1LRdo7d4Lc3vZmJpjLSnB7MwotakJGakDaGSF2Cm24o5QkgNUeMB45sENhBB6
pk6CLVS3PLAhetDnR+q+LAgMBAAE=
Balance = 20
```

10. Customer1 的 disconnected 訊息

```
parsed_message: {'request': 'close'}
address 'MEgCQQCW4GVfHAZTUJjAw7lRAQXDL1BGF9Exzk08vtJ5YvUSrvZ+JqsMbQtWTTjpXbNEQ1G
d6GR0Go+acBTBInIC7HTTAgMBAAE=' has disconnected to the miner.
```

11. Customer2 的 disconnected 訊息

```
parsed_message: {'request': 'close'}  
address 'MEgCQQCUvg1LRdo7d4Lc3vZmJpjLSnB7MwotakJGakDaGSF2Cm24o5QkgNUeMB45sENhBB6  
pk6CLVS3PlAhetDnR+q+LAgMBAAE=' has disconnected to the miner.
```

心得：這份作業對我來說很有挑戰性，在努力了好幾天後看到 miner 和 customer 之間真的連接起來並能做出互動，就覺得很有成就感。